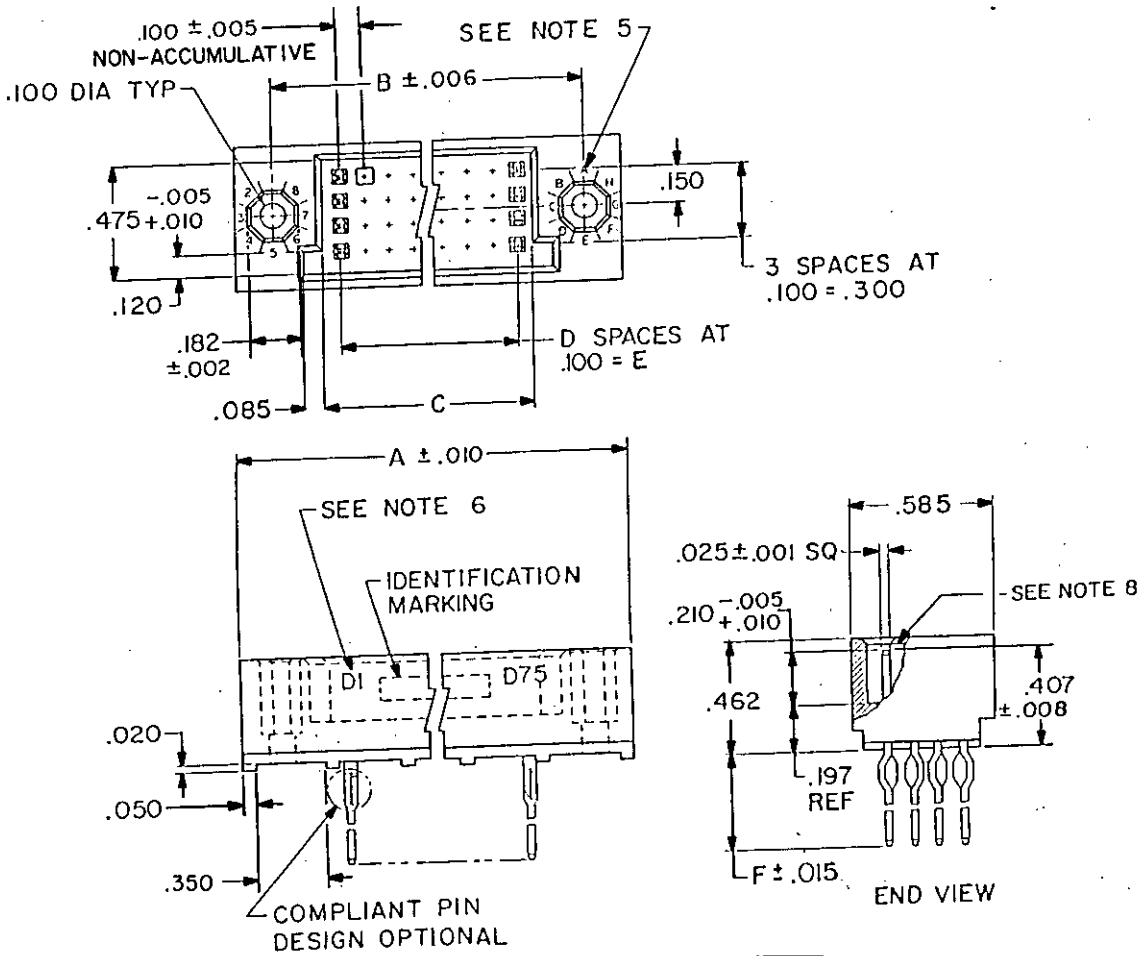
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 4-ROW PIN ASSEMBLIES, 100 THROUGH 684 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



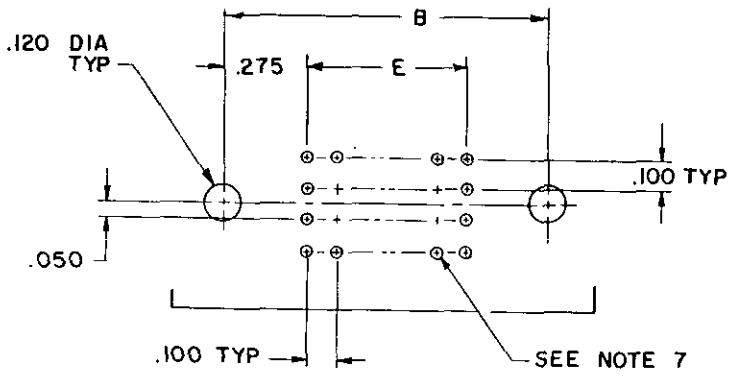
F ±.015		Type	Type of termination
Inches	mm		
.733	18.62	I	Complaint pin
.533	13.54	II	"
.328	8.33	III	"
.250	6.35	IV	"
.177	4.50	V	Solder post 1/
.190	4.83	VI	" 1/
.165	4.19	VII	" 1/
.140	3.56	VIII	" 1/
.120	3.05	IX	

1/ Solder post shall be a .036 (0.91 mm) maximum diameter/or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts.

FSC 5935
AMSC N/A

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Recommended circuit layout

F ±.015		Type	Type of termination
inches	mm		
.733	18.62	I	Compliant pin
.533	13.54	II	"
.328	8.33	III	"
.250	6.35	IV	"
.177	4.50	V	"
.190	4.83	VI	Solder post 1/
.165	4.19	VII	"
.140	3.56	VIII	"
.120	3.05	IX	"

1/ Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.050	1.27
.085	2.16
.100	2.54
.120	3.05
.150	3.81
.182	4.62
.190	4.83
.197	5.00
.210	5.33
.250	6.35
.275	6.98
.300	7.62
.350	8.89
.407	10.34
.462	11.73
.475	12.06
.533	13.54
.585	14.86

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/180.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and markings every ten positions stamped this surface.
7. Holes for solder type shall be .042-.050 (1.067-1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ±.0010 (1.15 ±0.0254 mm) drilled hole (1.15 mm drill) .001-.003 (0.025-0.076 mm) thick copper on wall, .0003 (0.0076 mm) min. tin lead plating, finished hole diameter .037-.043 (0.94-1.09 mm) after plating, .036-.043 (0.91-1.09 mm) after reflow, copper hardness (Knoop) 150 maximum.
8. Chamfers shall be .015 minimum at a 45° angle ±5 degrees (see figure 1, 2, or 3).

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts - Continued.

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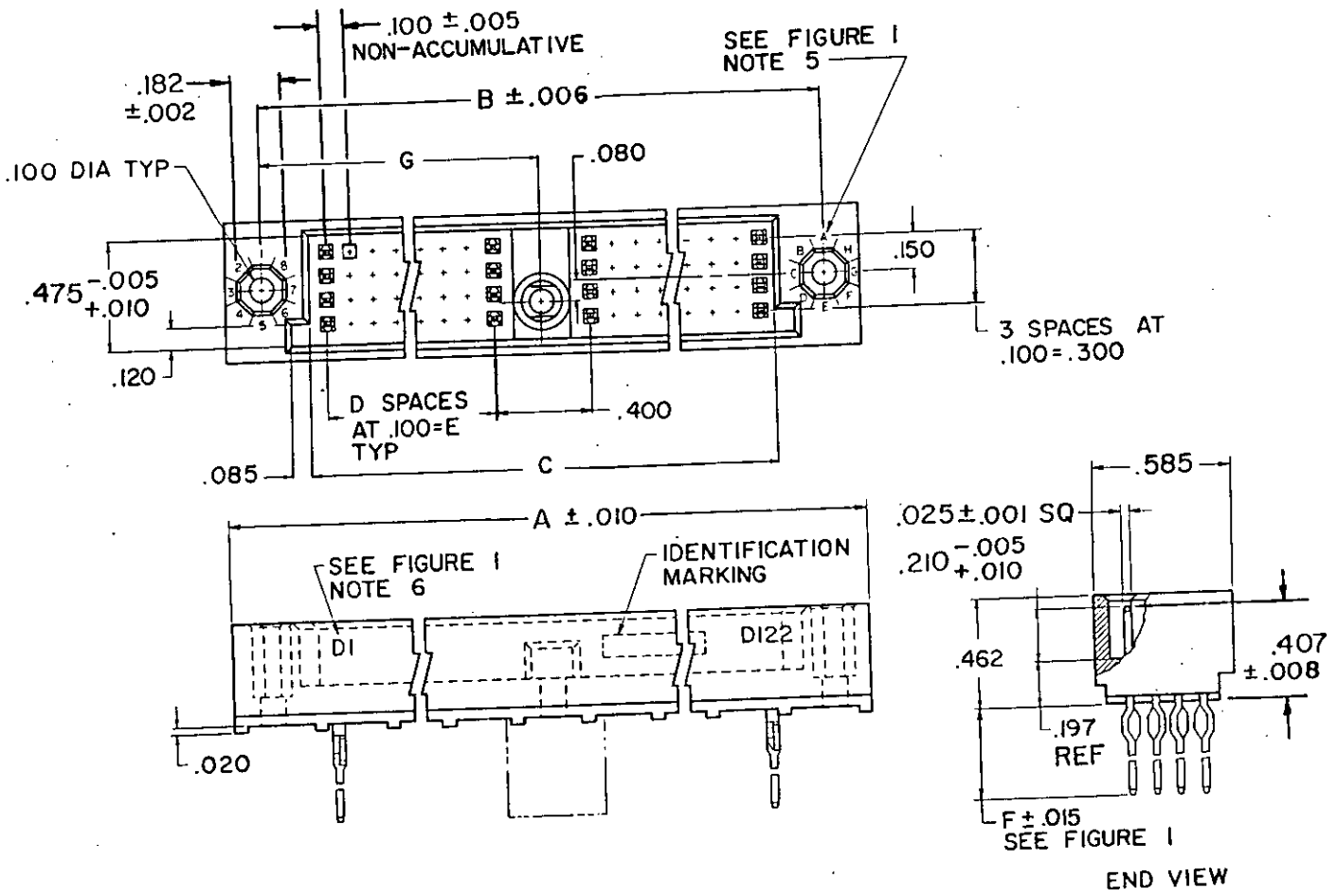
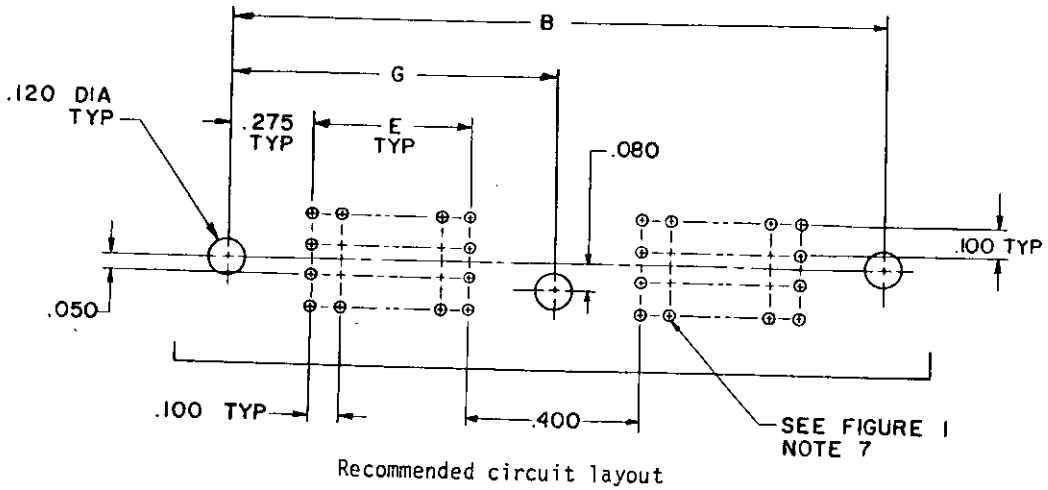


FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.050	1.27
.080	2.03
.085	2.16
.100	2.54
.120	3.05
.182	4.62
.197	5.00
.210	5.33
.275	6.98
.300	7.62
.400	10.16
.407	10.34
.462	11.73
.475	12.06
.585	14.86

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts - Continued.

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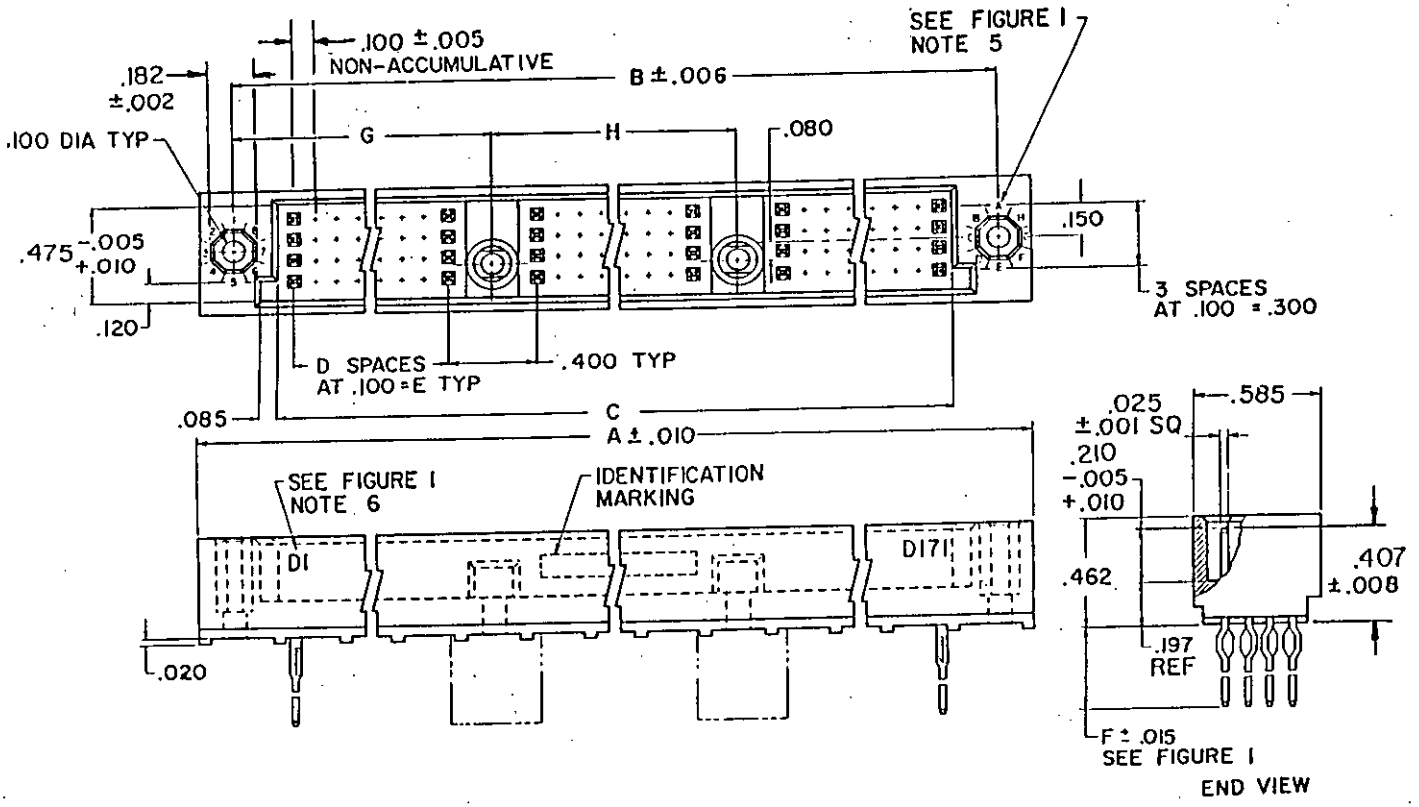
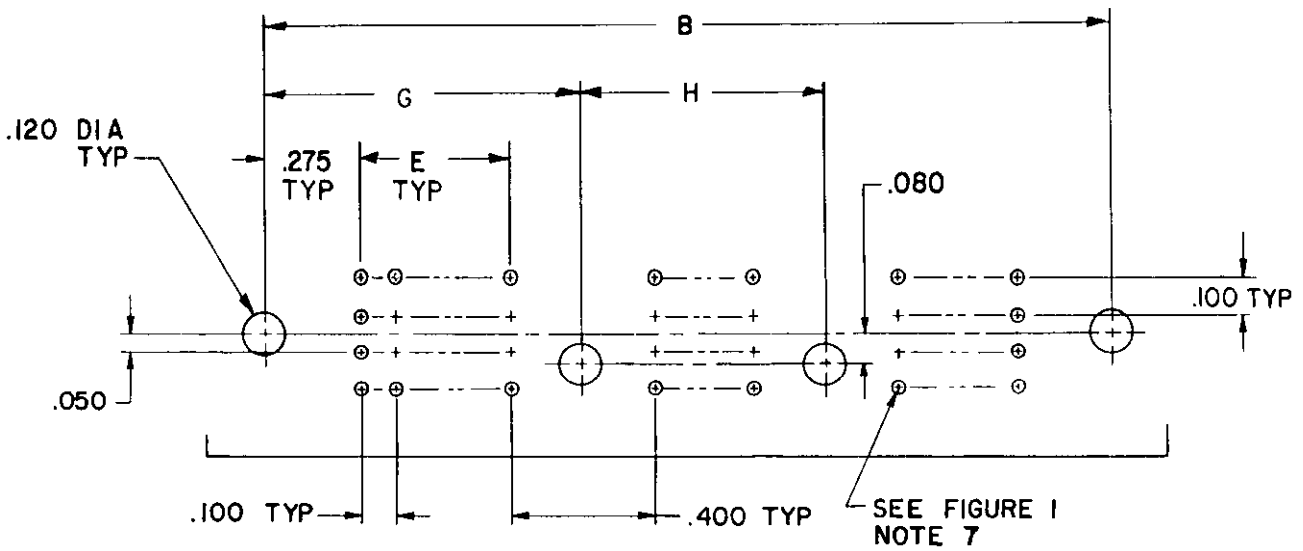


FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.050	1.27
.080	2.03
.085	2.16
.100	2.54
.120	3.05
.150	3.81
.182	4.62
.197	5.00
.210	5.33
.275	6.98
.300	7.62
.400	10.16
.407	10.34
.462	11.73
.475	12.06
.585	14.86

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01	100	1	I	3.250	2.950	2.550	24	2.400			1-25
02			II	(82.55)	(74.93)	(64.77)		(60.96)			
03			III								
04			IV								
05			V								
06			VI								
07			VII								
08			VIII								
09			IX								
10	120	1	I	3.750	3.450	3.050	29	2.900			1-30
11			II	(95.25)	(87.63)	(77.47)		(73.66)			
12			III								
13			IV								
14			V								
15			VI								
16			VII								
17			VIII								
18			IX								
19	128	1	I	3.950	3.650	3.250	31	3.100			1-32
20			II	(100.33)	(92.71)	(82.55)		(78.74)			
21			III								
22			IV								
23			V								
24			VI								
25			VII								
26			VIII								
27			IX								
28	140	1	I	4.250	3.950	3.550	34	3.400			1-35
29			II	(107.95)	(100.33)	(90.17)		(86.36)			
30			III								
31			IV								
32			V								
33			VI								
34			VII								
35			VIII								
36			IX								
37	160	1	I	4.750	4.450	4.050	39	3.900			1-40
38			II	(120.65)	(113.03)	(102.87)		(99.06)			
39			III								
40			IV								
41			V								
42			VI								
43			VII								
44			VIII								
45			IX								

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
46	180	1	I	5.250 (133.35)	4.950 (125.73)	4.550 (115.57)	44	4.400 (111.76)			1-45
47			II								
48			III								
49			IV								
50			V								
51			VI								
52			VII								
53			VIII								
54			IX								
55	200	1	I	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
56			II								
57			III								
58			IV								
59			V								
60			VI								
61			VII								
62			VIII								
63			IX								
64	220	1	I	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)			1-55
65			II								
66			III								
67			IV								
68			V								
69			VI								
70			VII								
71			VIII								
72			IX								
73	240	1	I	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
74			II								
75			III								
76			IV								
77			V								
78			VI								
79			VII								
80			VIII								
81			IX								
82	260		I	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)			1-65
83			II								
84			III								
85			IV								
86			V								
87			VI								
88			VII								
89			VIII								
90			IX								

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
91 92 93 94 95 96 97 98 99	268		I II III IV V VI VII VIII IX	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
100 101 102 103 104 105 106 107 108	280	1	I II III IV V VI VII VIII IX	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70
109 110 111 112 113 114 115 116 117	300	1	I II III IV V VI VII VIII IX	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)			1-75
118 119 120 121 122 123 124 125 126	320	2	I II III IV V VI VII VIII IX	9.050 (229.87)	8.750 (222.25)	8.350 (212.09)	39	3.900 (99.06)	4.375 (111.13)		1-80
127 128 129 130 131 132 133 134 135	344	2	I II III IV V VI VII VIII IX	9.650 (245.11)	9.350 (237.49)	8.950 (227.33)	42	4.200 (106.68)	4.675 (118.75)		1-86

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
136 137 138 139 140 141 142 143 144	368	2	I II III IV V VI VII VIII IX	10.250 (260.35)	9.950 (252.73)	9.550 (242.57)	45	4.500 (114.30)	4.975 (126.37)		1-92
145 146 147 148 149 150 151 152 153	392	2	I II III IV V VI VII VIII IX	10.850 (275.59)	10.550 (267.97)	10.150 (257.81)	48	4.800 (121.92)	5.275 (133.99)		1-98
154 155 156 157 158 159 160 161 162	416	2	I II III IV V VI VII VIII IX	11.450 (290.83)	11.150 (283.21)	10.750 (273.05)	51	5.100 (129.54)	5.575 (141.61)		1-104
163 164 165 166 167 168 169 170 171	440	2	I II III IV V VI VII VIII IX	12.050 (306.07)	11.750 (298.45)	11.350 (288.29)	54	5.400 (137.16)	5.875 (149.23)		1-110
172 173 174 175 176 177 178 179 180	464	2	I II III IV V VI VII VIII IX	12.650 (321.31)	12.350 (313.69)	11.950 (303.53)	57	5.700 (144.78)	6.175 (156.85)		1-116

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
181 182 183 184 185 186 187 188 189	488	2	I II III IV V VI VII VIII IX	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.47)		1-122
190 191 192 193 194 195 196 197 198	516	3	I II III IV V VI VII VIII IX	14.250 (361.95)	13.950 (354.33)	13.550 (344.17)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
199 200 201 202 203 204 205 206 207	540	3	I II III IV V VI VII VIII IX	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
208 209 210 211 212 213 214 215 216	564	3	I II III IV V VI VII VIII IX	15.450 (392.43)	15.150 (384.81)	14.750 (374.65)	46	4.600 (116.84)	5.075 (128.91)	5.000 (127.00)	1-141
217 218 219 220 221 222 223 224 225	588	3	I II III IV V VI VII VIII IX	16.050 (407.67)	15.750 (400.05)	15.350 (389.89)	48	4.800 (121.92)	5.275 (133.99)	5.200 (132.08)	1-147

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
226 227 228 229 230 231 232 233 234	612	3	I II III IV V VI VII VIII IX	16.650 (422.91)	16.350 (415.29)	15.950 (405.13)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
235 236 237 238 239 240 241 242 243	636	3	I II III IV V VI VII VIII IX	17.250 (438.15)	16.950 (430.53)	16.550 (420.37)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
244 245 246 247 248 249 250 251 252	660	3	I II III IV V VI VII VIII IX	17.850 (453.39)	17.550 (445.77)	17.150 (435.61)	54	5.400 (137.16)	5.875 (149.23)	5.900 (147.32)	1-165
253 254 255 256 257 258 259 260 261	684	3	I II III IV V VI VII VIII IX	18.450 (468.63)	18.150 (461.01)	17.750 (450.85)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

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TABLE II. Contact plating requirements.

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area, .170 inch minimum length, shall be gold over nickel in accordance with MIL-C-55302. Compliant pin area (contact type I, .748 inch minimum length; type II, .548 minimum length; type III, .343 inch minimum length; type IV, .265 inch minimum length; type V, .192 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302. Solder post area (contact type VI, .205 minimum length; type VII, .190 inch minimum length; type VIII, .165 inch minimum length; type IX, .135 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (86/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290,and shall meet the requirements of MIL-P-45209.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, 3, and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-C-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Compliant section shall meet the requirements of MIL-STD-2166, type III. For use with .093 minimum thick printed wiring boards. Sequence of tests and number of samples shall be in accordance with MIL-STD-2166, table III.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .094; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

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Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number: M55302/179 (and dash number from table I and type from table II).

Example: M55302/179-01A.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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